KNN Algorithm

CLASSIFICATION USING KNN

The k-nearest neighbors (kNN) algorithm determines the target value of a new data point or instance by comparing it with existing data points or instances that are closest to it.

The target values of the k-closest instances are aggregated taken at the target output for the new data point.

Consider the following data set

defaulter	student	balance	income
No	Yes	817.1804	12106.14
Yes	Yes	1486.998	17854.4
Yes	Yes	1102.435	17391.78
Yes	Yes	1233.446	12586.48
Yes	Yes	1276.685	19282.44
Yes	Yes	1289.246	13624.55
No	Yes	673.8124	17672.29
No	No	870.4124	17685.45
No	Yes	903.1768	15810.57
No	Yes	879.627	14898.31
unknown	Yes	1000	16900

For the given data set we will try to predict the value of the target column default are based on the other columns such as the income and balance.

for the given data set by creating a scatter plot we find that there is a balance of 1000 and an income of

16,900 which can be classified as defaulter or a non defaulter. in such case we shall use kNN to classify this instance.

Euclidean distance

In the KNN algorithm we identify the nearest neighbors based on euclidean distance.
the euclidean distance between 2 tuples- $X_1(x_{11},x_{12},x_{13},\ldots,x_{1n})$ and $X_2(x_{21},x_{22},x_{23},\ldots,x_{2n})$ Can be computed by Euclidean_dist(X_1, X_2)= $\sqrt{\sum_{i=1}^{n}(x1i - x2i)^2}$
The value of the euclidean distance for the above = 5787.154.
The difference in the values varies between every pair of randomly selected variables. this is a limitation of the euclidean distance. to eliminate this we normalize all the numerical attributes of the couples before they are computed.

Normalizing the data

Let us use the min Max normalization to normalize the data.

$$v' = \frac{v - minA}{maxA - minA}$$

It is difficult to classify the data as defaulter or a non defaulter because the count results in a tie. for any binary class cases such as yes/no odd value of K is preferred to avoid such tie situations.
on the other hand for multi classes(more than 2) the ties r classified by assigning a class at random or by

assigning the class that occurs most frequently.

Building KNN model using sklearn

Steps involved:

Step 1- loading the data

Step 2- feature engineering or Normalization

Step 3- splitting the data set into train and test data

Step 4- building the model

Step 5- evaluate model performance on train and test sets